Product Information
Acronal® ECO 612 ap

Polymer dispersion for low environmental impact near zero-low VOCs, low odor architectural coatings

Acronal® ECO 612 ap is a full acrylic copolymer developed for flat to low sheen architectural paints. Paint formulated with Acronal® ECO 612 ap meets stringent performance requirements while keeping VOC content near zero. Due to its excellent adhesion and block resistance, paint containing Acronal® ECO 612 ap is suitable for a wide range of surface types and applications. In addition, Acronal® ECO 612 ap showed excellent resistance to most common household stains

Chemical Nature:

Polymer dispersion of acrylic ester

Benefits

- Earlier reoccupation of painted space
- Minimal need for coalescing agent
- Excellent block resistance
- Good stain resistance
- Meets or exceeds VOC regulations paint

Features

- Low odor polymer dispersion
- Low MFFT
- APEO and formaldehyde free

Properties			
Product specification*	Solids content	%	49 ± 1
	pH value	pH	6.0 – 8.0
	Viscosity at 23 °C, shear rate 100^{-1} (DIN EN ISO 2555)	mPa·s	200 – 1000
Other properties of dispersions	Minimum film-forming temperature (ISO 2115)	°C	< 5
	Density (ISO 2811-1)	g/cm^3	approx. 1.04
	Resistance to frost	ºC	≤ 0
	Type of dispersion		anionic

*The aforementioned data shall constitute the agreed contractual quality of the product at the time of passing of risk. The data are controlled at regular intervals as part of our quality assurance program. Neither these data nor the properties of product specimens shall imply any legally binding guarantee of certain properties or of fitness for a specific
purpose. No liability of ours can be derived therefrom.

Note
The information submitted in this publication is based on our current knowledge and experience. In view of the many factors that may affect processing and application, these data do not relieve processors of the responsibility of carrying out their own tests and experiments; neither do they imply any legally binding assurance of certain properties or of suitability for a specific purpose. It is the responsibility of those to whom we supply our products to ensure that any proprietary rights and existing laws and legislation are observed.

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Applications

Areas of application

Acronal® ECO 612 ap is ideal for zero-low VOC and low odor paints. The dispersion allows you to formulate high quality matt to low sheen premium interior while meeting or exceeding VOC regulations. Due to its excellent washability, Acronal® ECO 612 ap based paint, allows most common household stains to be cleaned away with common detergents.

Processing

It is advisable to disperse the pigments and extenders with wetting and dispersing agents such as sodium polyacrylates or sodium polyphosphates, which are particularly appropriate for use as dispersing agents in low-odor coatings. It is only when products with very high viscosity are being mixed in low-speed mixers that Acronal® ECO 612 ap should be added together with the wetting and dispersing agents.

The nature and quantity of the pigments and fillers can be varied within wide limits. This primarily depends on the application for which the coating is intended, i. e. interior or exterior, factors such as the required gloss, flexibility and wet abrasion resistance, the type of substrate and the amount of binder required to disperse the pigments and fillers (i.e. their oil absorption).

Various thickeners can be added to emulsion paints in order to adjust their viscosity and workability. Cellulose ethers, polyacrylates, diurethane thickener (such as Rheovis® AS 1125) and bentonite can be used. The choice of thickener depends on whether the coating is expected to be free-flowing or more thixotropic.

No solvents need to be added to medium pigmented interior paints formulated with Acronal® ECO 612 ap to promote film formation.

Like all finely divided polymer dispersions, Acronal® ECO 612 ap has a tendency to foam. It is therefore necessary to add a commercial defoamer at the level recommended by the manufacturer. Trials should be carried out to test the effectiveness of the defoamer.

Although Acronal® ECO 612 ap itself is resistant to microorganisms in the form in which it is supplied, preservatives need to be added to products formulated with Acronal® ECO 612 ap to protect them from attack by microorganisms over long periods in storage. Trials should always be carried out to test the compatibility and efficacy of the preservatives.

Customers have to carry out their own trials when developing and processing products based on Acronal® ECO 612 ap. The compatibility of Acronal® ECO 612 ap with other ingredients of formulations, its effect on mixing processes and its adhesion on different substrates etc., are affected by a variety of factors which are too numerous for us to take into account in our own trials. This includes testing its stability by storing it at ca. 50 °C to confirm that its viscosity remains stable.

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Safety

General
The usual precautions for handling chemicals must be observed. These include the measures set out in the guidelines of the organisations responsible for safety at work, in particular, good ventilation and fume extraction at the workplace, care of the skin and the wearing of eye protection.

Safety Data Sheet
When using this product, the information and advice given in our Safety Data Sheet should be observed. Due attention should also be given to the precautions necessary for handling chemicals.

Labeling
According to all the data at our disposal, Acronal® ECO 612 ap does not need to be labelled as a dangerous substance or preparation as defined in the relevant local directives according to their current status.

Storage

Acronal® ECO 612 ap must not be allowed to come into contact during storage with metals or alloys that are susceptible to corrosion. It is important to ensure that containers are kept tightly sealed, and the headspace of bulk storage tanks must be kept saturated with water vapor. This product must not be exposed to high temperatures, and it must be protected from frost.

Acronal® ECO 612 ap has a shelf life of nine months at 10 – 30 °C, provided due attention is paid to the hygiene of tanks and storage facilities..

We would recommend treating this product with a biocide in order to prevent problems with micro organisms from occurring during storage and processing. Further details are given in our leaflet on “The handling and storage of polymer dispersions”.

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